INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science & Engineering

Branch :- CSE **Semester – Second** **Section- CSE-1/ CSE-2**

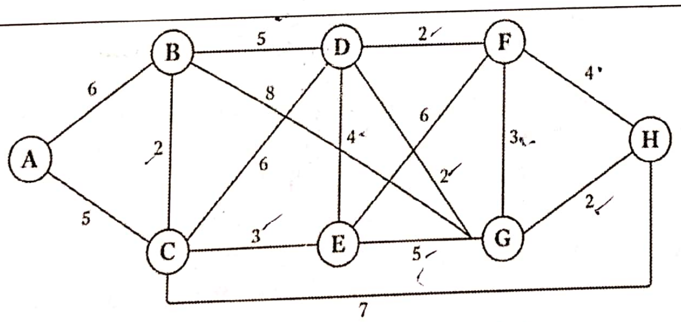
End-Term Examination, May – 2026

Course : B. Tech

Subject Code: - CSE-2004 Max. Marks: 60
Subject Name:- Data Structures and Algorithms Time: 03hr

NOTE: 1. All questions are compulsory.
2. While solving write question no. and its part clearly.
3. Solve questions in given sequence and all the sub parts should be at the same place.

S. No		Question	Marks
1.	(a)	Using the given In-order and Post-order Traversals, construct the corresponding Binary tree. (Show all intermediate steps clearly) **Inorder Traversal: H D I B J E K A L F M C N G** **Postorder Traversal: H I D J K E B L M F N G C A**	05
	(b)	**45, 30, 20, 60, 35, 55, 57, 70, 68, 10, 25, 8** Using the given sequence of numbers, construct an AVL Tree and count the no. of rotations. RL and LR rotations must be counted as two rotations. Show detailed steps for construction of the AVL Tree.	05
2.	(a)	Consider the following list of elements: **25, 38, 10, 48, 59, 18, 69, 81, 5, 13** Construct a Max Heap from the given elements using the Heapify method and Perform Heap Sort to arrange the elements in increasing order. Show every step clearly.	05
	(b)	Write a C/C++ program to implement a Circular Queue using an array. The program must include all the associated functions.	05
3.	(a)	Consider the following list of elements: **65, 34, 78, 12, 90, 23, 56, 11, 43, 89, 28, 50** Assume the first element (65) as the pivot. Perform Pass I of Quick Sort (Partitioning step) and write the final array after completion of the first pass.	05
	(b)	Write the Pseudocode of the Quick Sort Algorithm. And give a detailed analysis of its time complexity in the best and worst case.	05

4.	(a)	Given the following sequence of elements: **20, 10, 30, 5, 15, 25, 35, 2, 7, 12, 18, 22, 27, 32, 40, 50** Create a B-Tree of order 4 by inserting the elements one by one into the B-Tree. (Show all intermediate steps clearly)	05
	(b)	From the constructed B-Tree, Delete the following elements- 7, 30. Clearly show the intermediate steps and draw the final B-tree.	03
	(c)	Differentiate between a B-Tree and B+- Tree.	02
5.	(a)	Consider a hash table of size 11. The following hash functions are used: $\mathbf{h_1(k) = k \bmod 11}$ $\mathbf{h_2(k) = 7 - (k \bmod 7)}$ Insert the following keys into the hash table using Double Hashing and show the final hash table after all insertions. **21, 44, 56, 32, 18, 73, 11, 39, 50**	05
	(b)	Calculate the total no. of collisions and the total no. of probes.	03
	(c)	Write C/C++ code to Create a singly linked list by taking a sequence of integer values from the user. Then, Traverse the linked list and: • Insert all elements at odd positions into a Stack • Insert all elements at even positions into a Queue	02
6.	(a)	 Using Dijkstra's Algorithm, determine the shortest path from source vertex A to destination vertex H and find its cost. Show all intermediate steps and calculations clearly.	05
	(b)	Construct the Minimum Spanning Tree (MST) of the given graph using Prim's Algorithm (starting from vertex A) and find the cost of the MST.	03
	(c)	Define a Minimum Spanning Tree (MST) and State the key differences between Prim's Algorithm and Kruskal's Algorithm.	02

END